Linux 第一次实验

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实验环境：Ubuntu 26.04 LTS（虚拟机 192.168.219.128，用户 breeeze）
仓库：https://github.com/qingfeng-13/linux2026

1. Directory Operation 目录操作

```
breeeze@breeeze-VMware-Virtual-Platform:~$ echo 付来 B24040408
付来 B24040408
breeeze@breeeze-VMware-Virtual-Platform:~$ pwd
/home/breeeze
breeeze@breeeze-VMware-Virtual-Platform:~$ cd ~
breeeze@breeeze-VMware-Virtual-Platform:~$ mkdir 0408
mkdir: 408: File exists
breeeze@breeeze-VMware-Virtual-Platform:~$ cd 0408
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ mkdir cs
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ ls
cs
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ rmdir cs
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ ls
breeeze@breeeze-VMware-Virtual-Platform:~/0408$
```

2. File operation 文件操作

```
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ echo 付来 B24040408
付来 B24040408
breeeze@breeeze-VMware-Virtual-Platform:~/0408$ cd ~
breeeze@breeeze-VMware-Virtual-Platform:~$ touch fulai.txt
breeeze@breeeze-VMware-Virtual-Platform:~$ ls
0408       Music      Videos           fulai.txt      linklab.tar.2  vm_clean.sh      vm_sudo_test.sh
Desktop    Pictures   _vm              linklab        njupt          vm_clean2.sh     web
Documents  Public     archlab          linklab.tar    shots          vm_gui_clean.sh
Downloads  Templates  clean_result.txt  linklab.tar.1  snap           vm_sudo_clean.sh
breeeze@breeeze-VMware-Virtual-Platform:~$ echo "hello word">0408.txt
breeeze@breeeze-VMware-Virtual-Platform:~$ cat 0408.txt
hello word
breeeze@breeeze-VMware-Virtual-Platform:~$ cp fulai.txt 0408_copy.txt
breeeze@breeeze-VMware-Virtual-Platform:~$ mv fulai.txt 0408_newname.txt
breeeze@breeeze-VMware-Virtual-Platform:~$ ls
0408              Desktop    Pictures   _vm              linklab.tar    shots          vm_gui_clean.sh
0408.txt          Documents  Public     archlab          linklab.tar.1  snap           vm_sudo_clean.sh
0408_copy.txt     Downloads  Templates  clean_result.txt  linklab.tar.2  vm_clean.sh    vm_sudo_test.sh
0408_newname.txt  Music      Videos     linklab          njupt          vm_clean2.sh   web
breeeze@breeeze-VMware-Virtual-Platform:~$
```

3. User Management 用户管理

```
breeeze@breeeze-VMware-Virtual-Platform:~$ echo 付来 B24040408
付来 B24040408
breeeze@breeeze-VMware-Virtual-Platform:~$ whoami
breeeze
breeeze@breeeze-VMware-Virtual-Platform:~$ sudo su root
[sudo: authenticate] 密码:
root@breeeze-VMware-Virtual-Platform:/home/breeeze# groupadd student
root@breeeze-VMware-Virtual-Platform:/home/breeeze# whoami
root
root@breeeze-VMware-Virtual-Platform:/home/breeeze# adduser NY0408
fatal: "NY0408"ç»
                å·²ç»å ¨ã
root@breeeze-VMware-Virtual-Platform:/home/breeeze# cd ~
root@breeeze-VMware-Virtual-Platform:~# touch NY0408.txt
root@breeeze-VMware-Virtual-Platform:~# ls -l
total 8
-rw-r--r-- 1 root root    0 Oct 14 20:44 NY0408.txt
drwx------ 9 root root 4096 Apr 16 22:20 snap
-rw-r--r-- 1 root root    0 Oct  8 21:08 sudotest
-rw-r--r-- 1 root root 3104 Oct 14 10:39 vm-restore-backup-20261014-1039.tar.gz
root@breeeze-VMware-Virtual-Platform:~#
```

4. Permission management 权限管理

```
breeeze@breeeze-VMware-Virtual-Platform:~$ echo 付来 B24040408
付来 B24040408
breeeze@breeeze-VMware-Virtual-Platform:~$ touch demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-rw-rw-r-- 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod 777 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-rwxrwxrwx 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod a-x demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-rw-rw-rw- 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod go-w demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-rw-r--r-- 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod g+w demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-rw-rw-r-- 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod a= demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
---------- 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod +t demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
---------T 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod u+s demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
---S-----T 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod u+rx demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-r-s-----T 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$ chmod g+rws demo
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l demo
-r-srwS--T 1 breeeze breeeze 0 Oct 14 20:47 demo
breeeze@breeeze-VMware-Virtual-Platform:~$
```

5. Other management 其他管理

```
breeeze@breeeze-VMware-Virtual-Platform:~$ cd ~
breeeze@breeeze-VMware-Virtual-Platform:~$ mkdir others
breeeze@breeeze-VMware-Virtual-Platform:~$ cd others
breeeze@breeeze-VMware-Virtual-Platform:~/others$ touch file1.txt file2.txt
breeeze@breeeze-VMware-Virtual-Platform:~/others$ cd ~
breeeze@breeeze-VMware-Virtual-Platform:~$ tar -czvf backup.tar.gz others
others/
others/file2.txt
others/file1.txt
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l
total 112
drwxrwxr-x  2 breeeze breeeze 4096 Oct 14 20:27 0408
-rw-rw-r--  1 breeeze breeeze   11 Oct 14 20:40 0408.txt
-rw-rw-r--  1 breeeze breeeze    0 Oct 14 20:41 0408_copy.txt
-rw-rw-r--  1 breeeze breeeze    0 Oct 14 20:38 0408_newname.txt
drwxr-xr-x  2 breeeze breeeze 4096 Apr 23 16:55 Desktop
drwxr-xr-x  3 breeeze breeeze 4096 Apr 23 17:00 Documents
drwxr-xr-x  2 breeeze breeeze 4096 Oct  8 18:43 Downloads
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Music
drwxr-xr-x  3 breeeze breeeze 4096 Apr 23 17:22 Pictures
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Public
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Templates
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Videos
drwxrwxr-x  4 breeeze breeeze 4096 Oct 14 16:07 _vm
drwxrwxr-x  3 breeeze breeeze 4096 Oct  8 21:07 archlab
-rw-rw-r--  1 breeeze breeeze  157 Oct 15 19:04 backup.tar.gz
-rw-r--r--  1 breeeze breeeze 3676 Oct  8 21:08 clean_result.txt
-r-srwS--T  1 breeeze breeeze    0 Oct 14 20:47 demo
drwxrwxr-x  3 breeeze breeeze 4096 Jun 11 12:19 linklab
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar.1
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar.2
drwxrwxr-x  2 breeeze breeeze 4096 Sep 16 15:39 njupt
drwxrwxr-x  2 breeeze breeeze 4096 Oct 15 19:04 others
drwxr-xr-x  2 breeeze breeeze 4096 Oct 14 11:43 shots
drwx------ 10 breeeze breeeze 4096 Sep 23 17:06 snap
-rwxrw-rw-  1 breeeze breeeze 1876 Oct  8 20:10 vm_clean.sh
-rwxrw-rw-  1 breeeze breeeze 3420 Oct  8 21:08 vm_clean2.sh
-rwxrw-rw-  1 breeeze breeeze 3369 Oct  8 20:12 vm_gui_clean.sh
-rwxrw-rw-  1 breeeze breeeze 3354 Oct  8 21:07 vm_sudo_clean.sh
-rwxrw-rw-  1 breeeze breeeze 1185 Oct  8 21:08 vm_sudo_test.sh
drwxrwxr-x  4 breeeze breeeze 4096 Oct 14 11:45 web
breeeze@breeeze-VMware-Virtual-Platform:~$ tar -xzvf backup.tar.gz
others/
others/file2.txt
others/file1.txt
```

```
breeeze@breeeze-VMware-Virtual-Platform:~$ ls -l
total 112
drwxrwxr-x  2 breeeze breeeze 4096 Oct 14 20:27 0408
-rw-rw-r--  1 breeeze breeeze   11 Oct 14 20:40 0408.txt
-rw-rw-r--  1 breeeze breeeze    0 Oct 14 20:41 0408_copy.txt
-rw-rw-r--  1 breeeze breeeze    0 Oct 14 20:38 0408_newname.txt
drwxr-xr-x  2 breeeze breeeze 4096 Apr 23 16:55 Desktop
drwxr-xr-x  3 breeeze breeeze 4096 Apr 23 17:00 Documents
drwxr-xr-x  2 breeeze breeeze 4096 Oct  8 18:43 Downloads
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Music
drwxr-xr-x  3 breeeze breeeze 4096 Apr 23 17:22 Pictures
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Public
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Templates
drwxr-xr-x  2 breeeze breeeze 4096 Apr 16 18:48 Videos
drwxrwxr-x  4 breeeze breeeze 4096 Oct 14 16:07 _vm
drwxrwxr-x  3 breeeze breeeze 4096 Oct  8 21:07 archlab
-rw-rw-r--  1 breeeze breeeze  157 Oct 15 19:04 backup.tar.gz
-rw-r--r--  1 breeeze breeeze 3676 Oct  8 21:08 clean_result.txt
-r-srwS--T  1 breeeze breeeze    0 Oct 14 20:47 demo
drwxrwxr-x  3 breeeze breeeze 4096 Jun 11 12:19 linklab
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar.1
-rw-rw-r--  1 breeeze breeeze 1704 May 30  2024 linklab.tar.2
drwxrwxr-x  2 breeeze breeeze 4096 Sep 16 15:39 njupt
drwxrwxr-x  2 breeeze breeeze 4096 Oct 15 19:04 others
drwxr-xr-x  2 breeeze breeeze 4096 Oct 14 11:43 shots
drwx------ 10 breeeze breeeze 4096 Sep 23 17:06 snap
-rwxrw-rw-  1 breeeze breeeze 1876 Oct  8 20:10 vm_clean.sh
-rwxrw-rw-  1 breeeze breeeze 3420 Oct  8 21:08 vm_clean2.sh
-rwxrw-rw-  1 breeeze breeeze 3369 Oct  8 20:12 vm_gui_clean.sh
-rwxrw-rw-  1 breeeze breeeze 3354 Oct  8 21:07 vm_sudo_clean.sh
-rwxrw-rw-  1 breeeze breeeze 1185 Oct  8 21:08 vm_sudo_test.sh
drwxrwxr-x  4 breeeze breeeze 4096 Oct 14 11:45 web
breeeze@breeeze-VMware-Virtual-Platform:~$
```

```
breeeze@breeeze-VMware-Virtual-Platform:~$ cat > textfile << EOF
no, I study in njupt. this line ends with 00
this line ends with .00
I am 25 years old.
He is 15.
It is 16.00
wy is the short of my name.
EOF
breeeze@breeeze-VMware-Virtual-Platform:~$ grep '^n' textfile
no, I study in njupt. this line ends with 00
breeeze@breeeze-VMware-Virtual-Platform:~$ grep '\.00$' textfile
this line ends with .00
It is 16.00
breeeze@breeeze-VMware-Virtual-Platform:~$ grep '5\..' textfile
breeeze@breeeze-VMware-Virtual-Platform:~$ grep '^[wy]' textfile
wy is the short of my name.
```

SSH 服务器

```
PS C:\Users\38479> ssh breeeze@192.168.219.128
breeeze@192.168.219.128's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

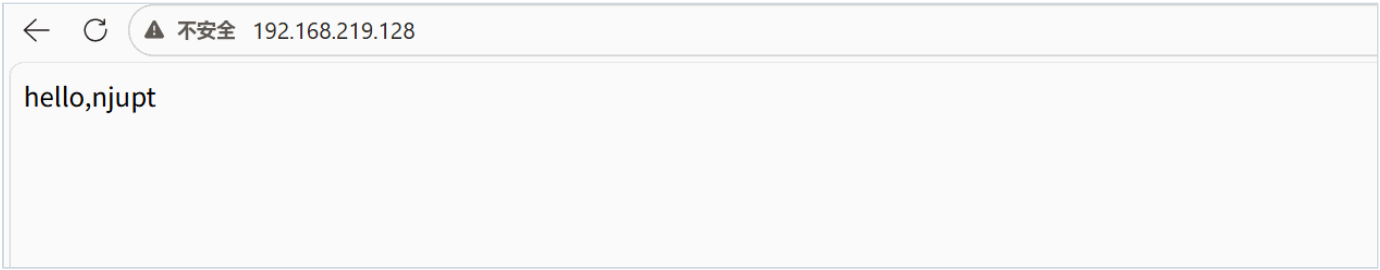
扩展安全维护（ESM）Applications 未启用。

15 更新可以立即应用。
这些更新中有 12 个是标准安全更新。
要查看这些附加更新，请运行：apt list --upgradable

16 个额外的安全更新可以通过 ESM Apps 来获取安装。
可通过以下途径了解如何启用 ESM Apps：at https://ubuntu.com/esm

Last login: Fri Sep  4 21:03:14 2026 from 192.168.219.1
breeeze@breeeze-VMware-Virtual-Platform:~$ pwd
/home/breeeze
breeeze@breeeze-VMware-Virtual-Platform:~$ ls
0408              Music      backup.tar.gz     njupt         vm_gui_clean.sh
0408.txt          Pictures   clean_result.txt  others        vm_sudo_clean.sh
0408_copy.txt     Public     demo              shots         vm_sudo_test.sh
0408_newname.txt  Templates  linklab           snap          web
Desktop           Videos     linklab.tar       textfile
Documents         _vm        linklab.tar.1     vm_clean.sh
Downloads         archlab    linklab.tar.2     vm_clean2.sh
breeeze@breeeze-VMware-Virtual-Platform:~$ mkdir njupt
mkdir: njupt: File exists
breeeze@breeeze-VMware-Virtual-Platform:~$ cd njupt
breeeze@breeeze-VMware-Virtual-Platform:~/njupt$ touch hello.txt
breeeze@breeeze-VMware-Virtual-Platform:~/njupt$ ls
hello.txt
breeeze@breeeze-VMware-Virtual-Platform:~/njupt$ echo "I love njupt" > hello.txt
breeeze@breeeze-VMware-Virtual-Platform:~/njupt$ cat hello.txt
I love njupt
breeeze@breeeze-VMware-Virtual-Platform:~/njupt$
```

Nginx 服务器



MySQL 服务器

```
breeeze@breeeze-VMware-Virtual-Platform:~$ sudo su root
root@breeeze-VMware-Virtual-Platform:/home/breeeze# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.4.11-0ubuntu0.26.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

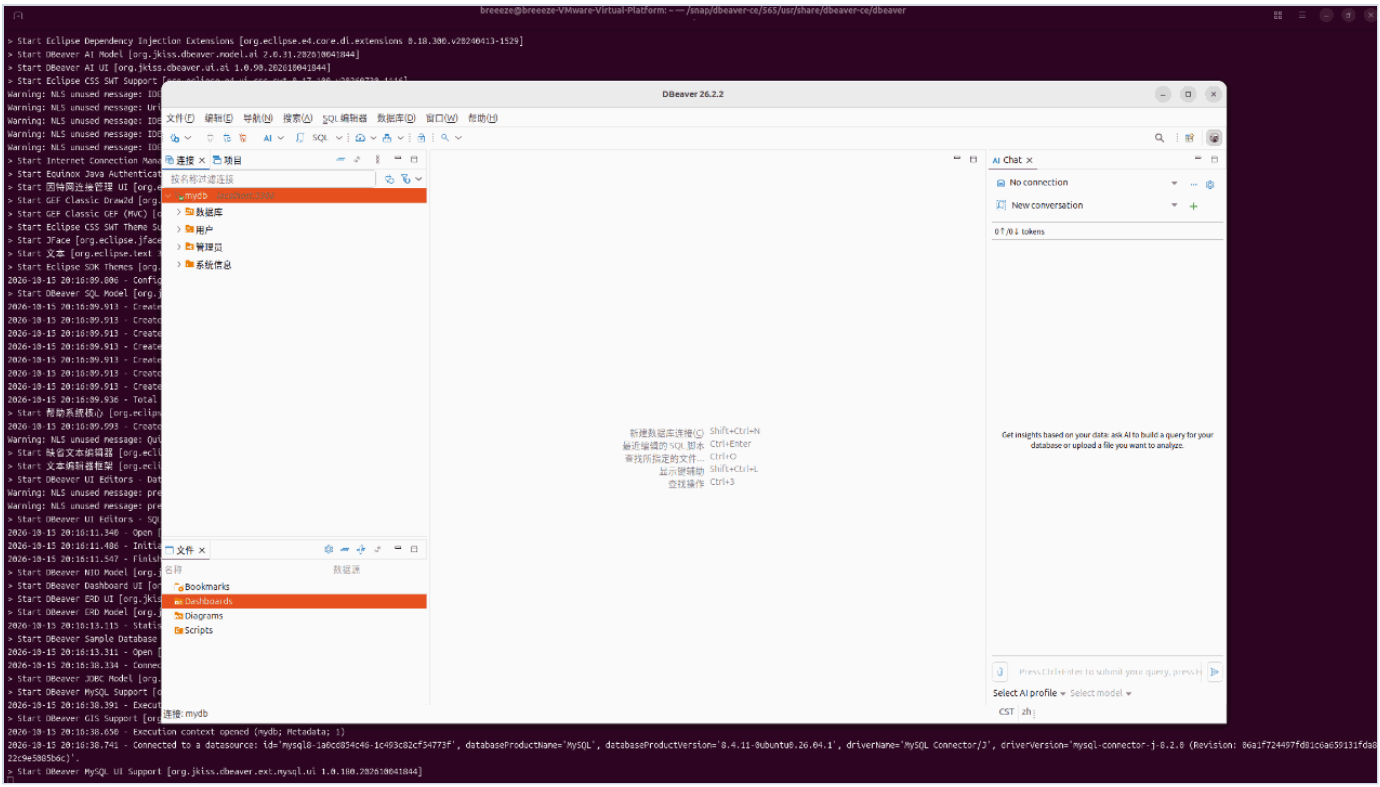
mysql> CREATE DATABASE mydb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE USER 'myuser'@'localhost' IDENTIFIED BY '111111';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON mydb.* TO 'myuser'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mydb               |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```


DBeaver 26.2.2
文件(F) 编辑(E) 导航(N) 搜索(A) SQL 编辑器 数据库(D) 窗口(W) 帮助(H)
连接
项目
按名称过滤连接
mydb
数据库
用户
管理员
系统信息
AI Chat
No connection
New conversation
0 ↑ /0 ↓ tokens
Get insights based on your data: ask AI to build a query for your database or upload a file you want to analyze.
Select AI profile
Select model
新建数据库连接(C) Shift+Ctrl+N
最近编辑的 SQL 脚本 Ctrl+Enter
查找所指定的文件... Ctrl+O
显示键辅助 Shift+Ctrl+L
查找操作 Ctrl+3
文件
名称
数据源
Bookmarks
Dashboards
Diagrams
Scripts
连接: mydb